

## Long-term shifts in the communities of odonata: effect of chance or climate change?

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**Abstract.** Global climate change has been causing growing concern among conservationists for its strong implications on biodiversity alteration and loss at different levels of organization. Dragonflies and damselflies (order Odonata) occur in habitats threatened by global warming, thus they represent an ideal model organism to study the correlation patterns of climate change with taxonomic composition and the ecological functioning of communities. We carried out climate and diachronic faunistic analyses of Odonata community changes in three countries (Tunisia, Mauritania, Sweden) to test if the patterns uncovered for single assemblages as a response to local climate change may resist to the generalization across regions and latitudes. Clear climate warming occurred in the analysed regions during the last five decades. We found three main patterns of diachronic shifts in Odonata assemblage species composition based on correlative evidence: i) Generalists are likely advantaged from warming processes that cause the loss of specific habitats (i.e. temporary wetlands, cool lentic waters) and the formation of new or altered habitats suitable for pioneer species (i.e. warm and intermittent pools), whereas specialists are more likely to go toward local extinctions; ii) In Tunisia and Sweden new colonizers expanded northward from their southern distributions; iii) The Odonata communities inhabiting lentic waters are more prone to show species turnover than communities from standing waters. Our results provide new insights on the possible impact of climate change on Odonata fauna from large areas (i.e. countries) at different latitudes and represent an attempt of a generalization of the effects of climate change on Odonata range shifts and expansions. Despite that Odonata global assessment of conservation status has been completed, insufficient information is available to robustly assess all the main threats affecting their status, and extensive new field surveys are required to test if major changes in fauna composition have occurred during the last decades.

**Key Words:** Climate change, Odonata, global warming, local extinction, community shifts.

### Introduction

Global climate change has been causing a growing concern among the international community of conservationists (Hannah et al. 2002a, 2002b, Arribas et al. 2017), especially during the last three decades that have seen an unprecedented rise in the speed of the meteorological and temperature changes (Meyer 2014).

During the last two decades, a growing body of studies on disparate taxa, communities and ecosystems highlighted these climate changes and invoked speculations on whether strong implications of “global warming” may affect biodiversity at different levels of organization, possibly even driving certain species towards extinction (Thomas et al. 2004). However, change in phenology and life cycles have been documented (Todd et al. 2010, Rugiero et al. 2013) as well as range extension/contractions (Parmesan & Yohe 2003, Walther et al. 2010). Among the climate change-related effects on phenological changes of natural animal populations, we may recognize earlier onset appearances (e.g. British butterflies: Roy & Sparks 2000, amphibians: Todd et al. 2010), general changes in activity periods (e.g. dragonflies: Ott 2001, snakes: Rugiero et al. 2013, Capula et al. 2014), northward expansion (Parmesan & Yohe 2003), disruptive migration patterns (e.g. overwintering of migratory water birds along the North Sea coast rather than the milder western seaboard of Britain; Robinson et al. 2005), microevolutionary and local adaptations (e.g. niche expansion of a butterfly; Davies et al. 2006), local extinction (e.g. butterflies in the southern parts of their geographical range; Hill et al. 2002), and changes in the community structure (e.g. dragonflies: Ott 2007; birds: Devictor et al. 2008; butterflies: Van Swaay et al. 2010).

Uncovering climate change-related alterations in natural

communities would require a long-term longitudinal approach, that is rather uncommon in modern ecological studies. Thus, we would ideally need to find appropriate surrogates that would give insights into a complicated scenario. In this regard, faunistic reports (checklist of species) carried out in different spatial-temporal contexts may convey noteworthy information on long term trends by comparing the different species pools that were observed at each sampling time and correlating the eventual differences with the climate change.

Dragonflies and damselflies (order Odonata) represent an ideal model taxon to investigate the correlation patterns of climate change with taxonomic composition and ecological functioning of communities. Indeed, for this animal group the species identification can be easily done at sight for most species (thus making the fieldwork feasible and straightforward) (Carle 1979) and dragonflies have been subjected to a long history of scientific research, including extensive amateur recording, with an outstanding historical faunistic database as a result (Corbet 2004, Kalkman et al. 2008, Kalkman 2018, Abbott 2019).

The response of Odonata to climate change (i.e. higher temperatures during summertime and warmer winters) would include alteration in phenology and morphology, range expansion, invasion of tropically distributed species, local extinction of cold environment-adapted species (Ott 2001, Hassall et al. 2007, 2008). For instance, range increases can be due to heavy storms that can, in turn, lead to long distance drifts of individuals with new areas being readily occupied by species; on the other side, desiccation of water bodies due to the increasing temperature may lead to the whole disappearance of local communities (Ott 2001). Moreover, the dragonfly community can be altered by variation of the water level by favouring generalist species (Ott 2007,

2010).

In this paper, we carry out diachronic analyses of Odonata community changes worldwide (based on the data availability), to uncover if the patterns already highlighted as responses to local climate change by distinct dragonfly assemblages (Ott 2010) may resist to the generalization across regions and over different latitudes.

## Material and Methods

### Literature research

We surfed Google Scholar, on 15 July 2017, to search for scientific papers with the following keywords: “Odonata faunistic survey” (total number of records: 2.370), “effects of global warming on Odonata” (total number of records 4.350), “dragonfly responses to climate change” (total number of records: 15.800), “Odonata change distribution” (total number of records: 23.700).

Among all records, we focused on checklist papers that were carried out in the same area but in different times, or across time, deriving from effective sampling activity. In this way, we avoid considering data deriving from amateur or museum collections. A minimum of 10-year time-span between two adjacent checklists from the same place was considered for our analyses, as this temporal distance is sufficient to observe a response by Odonata communities to the local climatic change (Flenner & Sahlén 2008).

We selected for further analyses only three study areas that fulfilled all the above-mentioned criteria: Tunisia (Dumont 1977, Jödicke et al. 2000), Mauritania (Dumont 1976, 1978, Ferreira et al. 2011) and Sweden (Flenner & Sahlén 2008).

Tunisia: The area surveyed by Jödicke et al. (2000), contained most of Dumont (1977) sites, and its range goes from 37°19' North to 32°57' South. Also, we consulted Gadeau de Kerville (1908) and Campion (1914) as a further source of information.

Mauritania: Ferreira et al. (2011) included a comparison with Dumont (1976, 1978) by providing accurate checklists: for every species recorded in Ferreira et al. (2011), it is pointed out if it has been sampled even by Dumont (1976, 1978) and whether each of Dumont's species were also included in their sample. The latitude range of Ferreira et al. (2011) sampling ranged from 25° 13,6' N to 14°45,3' N.

Sweden: the study by Flenner and Sahlén (2008) consists of a comparison of datasets at 34 lakes resampled at about 10 years distance. This paper includes also a climate analysis for the selected area over 10 years.

For the three countries, we gathered climate data from the website <http://sdwebx.worldbank.org/climateportal/index.cfm>, downloading average monthly temperature (°C) and rainfall (mm) for 50 years (1965-2015). All the data collection periods represented in the studies considered for our analyses were carried out in this time gap.

### Ecological analysis of the species

For each area, we built a reiterative table using the categories “Extinct” and “New” that included the species that disappeared from a given area in respect to older samplings or that were newly recorded in following surveys, respectively. We also depicted the ecological space of the relevant species (i.e., distribution, habitat preferences, phenology, oviposition strategy, etc.) by consulting reference publications on Odonata biology (i.e. Askew 1988, Silsby 2001, Corbet 2004, Paulson 2009, 2012). This was done to explore whether ecological key traits are shared across species showing the same pattern of extinction/colonization. Eventually, we assigned each relevant species into three categories: (i) Geographic area; (ii) Habitat; and (iii) Ecology. As for the geographic region, we simply indicated the country where the species was recorded (Tunisia, Mauritania or Sweden). The Habitat category included two main environment types, Lentic vs Lotic habitats, with lentic category containing also species capable

of living in the lotic habitat. The Ecology category assigned to the Generalist type (Gen) those species capable to survive in a wide range of habitat requirements as for vegetation (absent, low, abundant etc.) or temperature/water quality (i.e. permanent or temporary water bodies), and to the Specialist type (Sp) those species that require abundant vegetation or specific water regime/quality (i.e. cool, well-oxygenated water, specific plants requirements etc. Corbet 2004, Paulson 2009, 2012).

### Statistical analyses

Climate data for Tunisia, Mauritania and Sweden were analysed by means of the linear regression between the time and the average annual temperature and rainfall.

The pattern of Odonata species extinction/colonization was analysed by means of the log-linear analysis (James & McCulloch 1990), including as design variables the Area (Tunisia, Mauritania, Sweden), the Habitat type (lentic vs lotic), and the Ecology (Gen vs Sp) and their interactions as the effects on the response variable (i.e. the frequencies of extinct [Extinct] or newly discovered [New] species). We built a full model with all factorial interaction among design variables and we performed tests of Marginal and Partial Association on the models to uncover relationships between design variables and the response variable (James & McCulloch 1990).

All the analyses were done by using Statistica software (version 8.0; Statsoft) with alpha set at 0.05.

## Results

### Climate analyses.

Overall, the variation of the average temperatures showed the same pattern for all the selected areas, with a significant increase over the last 50 years (Tunisia:  $R = 0.81$ ,  $p < 0.0001$ ; Mauritania:  $R = 0.73$ ,  $p < 0.0001$ ; Sweden:  $R = 0.54$ ,  $p < 0.0001$ ). Annual precipitation did not show a significant trend for Tunisia ( $R = 0.05$ ,  $p = 0.68$ ) and Mauritania ( $R = 0.003$ ,  $p = 0.1$ ), while for Sweden we found a significant increase of average annual rainfall over the considered period ( $R = 0.35$ ,  $p = 0.011$ ).

### Pattern of extinction/colonization

The number of species that underwent extinction or colonized new areas is summarized in Table 1. From the log-linear analysis, we retained only the two models that were statistically significant with a consensus between partial and marginal associations (see Table 2). The first one contains only the Habitat variable, showing that overall the events of species extinction/colonization in lentic habitats (16/13) occurred more frequently than in lotic habitats (9/3), but the extinction/colonization ratio is similar in both habitat types (Table 2). The second model included the interaction between Ecology and the frequency of extinct/new species, with specialist species being more prone to undergo extinction and generalist species being more abundant among new colonizers, respectively.

## Discussion

Our analyses on climate confirmed the expected increase of average temperatures over the last decades in the selected areas, and it is assumed that this trend likely had and still has profound effect in shaping fauna compositions in terms of the species' turnover (Hassal & Thompson 2008, Fenoglio

Table 1. List of relevant species, their category of change, ecology, habitat and country. “Extinct”: the species that disappeared from a given area in respect to older samplings; “New”: species newly recorded in a given area in respect to older samplings; Sp = specialist; Gen = generalist.

| Country    | Species Category | Species                                  | Ecology | Habitat |
|------------|------------------|------------------------------------------|---------|---------|
| Tunisia    | Extinct          | <i>Lestes sponsa</i>                     | Sp      | Lentic  |
|            |                  | <i>Coenagrion puella kocheri</i>         | Sp      | Lentic  |
|            |                  | <i>Onychogomphus uncatus</i>             | Sp      | Lotic   |
|            |                  | <i>Sympetrum sanguineum</i>              | Gen     | Lentic  |
|            |                  | <i>Coenagrion mercuriale</i>             | Sp      | Lotic   |
|            |                  | <i>Boyeria irene</i>                     | Sp      | Lotic   |
|            | New              | <i>Erythromma viridulum</i>              | Gen     | Lentic  |
|            |                  | <i>Enallagma deserti</i>                 | Gen     | Lentic  |
|            |                  | <i>Orthetrum sabina</i>                  | Gen     | Lotic   |
|            |                  | <i>Orthetrum trinacria</i>               | Gen     | Lentic  |
|            |                  | <i>Brachythemis leucosticta</i>          | Gen     | Lentic  |
|            |                  | <i>Diplacodes lefebvrii</i>              | Sp      | Lentic  |
|            |                  | <i>Trithemis kirbyi ardens</i>           | Sp      | Lotic   |
|            |                  | <i>Pantala flavescens</i>                | Gen     | Lentic  |
|            |                  | <i>Selysiothemis nigra</i>               | Gen     | Lentic  |
|            |                  | <i>Urothemis edwardsii</i>               | Gen     | Lotic   |
| Mauritania | Extinct          | <i>Palpopleura deceptor</i>              | Gen     | Lentic  |
|            |                  | <i>Lestes pallidus</i>                   | Gen     | Lotic   |
|            |                  | <i>Agriocnemis zerajica</i>              | Sp      | Lotic   |
|            |                  | <i>Pseudagrion hamoni</i>                | Gen     | Lotic   |
|            |                  | <i>Anax imperator</i>                    | Gen     | Lentic  |
|            |                  | <i>Acisoma panorpoides ascalaphoides</i> | Sp      | Lentic  |
|            |                  | <i>Orthetrum chrysostigma</i>            | Gen     | Lotic   |
|            |                  | <i>Tramea basilaris</i>                  | Gen     | Lentic  |
|            | New              | <i>Urothemis edwardsii</i>               | Gen     | Lentic  |
|            |                  | <i>Sympetrum fonscolombii</i>            | Gen     | Lentic  |
|            |                  | <i>Trithemis annulata</i>                | Gen     | Lentic  |
| Sweden     | Extinct          | <i>Aeshna caerulea</i>                   | Sp      | Lentic  |
|            |                  | <i>Aeshna viridis</i>                    | Sp      | Lentic  |
|            |                  | <i>Coenagrion johanssoni</i>             | Sp      | Lentic  |
|            |                  | <i>Somatochlora metallica</i>            | Sp      | Lentic  |
|            |                  | <i>Coenagrion armatum</i>                | Sp      | Lentic  |
|            |                  | <i>Epithea bimaculata</i>                | Sp      | Lentic  |
|            |                  | <i>Coenagrion lunulatum</i>              | Sp      | Lentic  |
|            |                  | <i>Sympetrum flaveolum</i>               | Sp      | Lentic  |
|            |                  | <i>Lestes dryas</i>                      | Sp      | Lentic  |
|            |                  | <i>Sympetrum striolatum</i>              | Gen     | Lentic  |
|            | New              | <i>Aeshna cyanea</i>                     | Gen     | Lentic  |
|            |                  | <i>Aeshna mixta</i>                      | Gen     | Lentic  |
|            |                  | <i>Leucorrhinia caudalis</i>             | Gen     | Lentic  |
|            |                  | <i>Sympecma fusca</i>                    | Sp      | Lentic  |

et al. 2010, Ott & Samways 2010). Indeed, although there was merely correlational evidence for the positive relationship between climate change and Odonata fauna qualitative variations, we can speculate that climate change in some way is at least in part behind the observed fauna turnover.

Our study showed that among all the species considered (extinct and new colonizers), the most that showed a response to environmental variation over the considered periods are adapted to lentic habitats. Although we cannot exclude that this may emerge as a casual sampling or data property, it is interesting that, in general, lentic waters (and the species adapted to) seem more responsive to climate change. Indeed, climate changes enhance both new colonisations that exploit the emergence of new or vacant niches left empty, and local extinctions following the loss of sensitive

habitats (i.e. Mediterranean temporary ponds; Zacharias & Zamparas 2010).

The main pattern emerging from our analyses is the interaction between the ecology of the species and their response to the environmental variation (extinction/colonization). Climate change, likely, promoted the colonization of new generalist species and facilitated local extinctions of specialist species in the considered areas. Indeed, a clear difference in the response of the two ecological categories (specialist vs. generalist) to climate changes emerged. It is well recognized that climate changes directly act on habitats (Roshie et al. 2001, Parmesan & Yohe 2003); therefore, the observed pattern can be explained considering that the greater plasticity of generalist species allowed them to survive in unstable and unpredictable habitats, as they are

Table 2. Tests of all marginal and partial associations of factors (response and design variables). 1: Frequencies of extinct or newly discovered species; 2: Habitat (lentic vs lotic); 3: Ecology (Generalist vs Specialist); 4: Area (Tunisia, Mauritania, Sweden). Df= Degree of freedom.

| Model     | Df       | Partial Association |              | Marginal Association |              |
|-----------|----------|---------------------|--------------|----------------------|--------------|
|           |          | Chi-sqr.            | p            | Chi-sqr              | p            |
| 1         | 1        | 1.536               | 0.215        | 1.536                | 0.215        |
| <b>2</b>  | <b>1</b> | <b>5.550</b>        | <b>0.018</b> | <b>5.550</b>         | <b>0.018</b> |
| 3         | 1        | 0.160               | 0.680        | 0.160                | 0.680        |
| 4         | 2        | 0.726               | 0.695        | 0.726                | 0.695        |
| 12        | 1        | 0.426               | 0.514        | 0.760                | 0.383        |
| <b>13</b> | <b>1</b> | <b>7.073</b>        | <b>0.008</b> | <b>6.141</b>         | <b>0.013</b> |
| 14        | 2        | 6.462               | 0.039        | 4.705                | 0.095        |
| 23        | 1        | 1.008               | 0.315        | 0.769                | 0.380        |
| 24        | 2        | 2.806               | 0.246        | 1.743                | 0.418        |
| 34        | 2        | 7.797               | 0.020        | 5.466                | 0.065        |
| 123       | 1        | 0.067               | 0.796        | 0.444                | 0.505        |
| 124       | 2        | 0.142               | 0.931        | 0.040                | 0.975        |
| 134       | 2        | 1.691               | 0.429        | 1.651                | 0.438        |
| 234       | 2        | 0.164               | 0.921        | 0.268                | 0.875        |

not necessarily linked to specific environmental features (habitat type, temperature, vegetation cover, etc.) (Devictor *et al.* 2008). Moreover, what emerged at a regional scale for three countries is coherent with the worldwide decline of specialist species pattern in which the climate change can be considered among the main drivers (Clavel *et al.* 2011). These trends of response were confirmed by the ecological traits analysed for the species that underwent local extinction/colonization events during the time between the diachronic samplings. For instance, in Tunisia, the local extinction of “European” species and the arrival of sub-tropical taxa determined a northward distribution shift for both species’ categories, probably linked to the raising of temperature, as demonstrated for other animal groups (Brommer 2004, Mitikka *et al.* 2008). Moreover, most of the local extinct species were associated with mesophilic habitat (lentic waters surrounded by dense vegetation and/or woods). Conversely, most of the new-colonizer species were generalist and some were from “sub-desertic” areas (Askew 1988, Corbet 2004). These patterns for Tunisia, likely, revealed that climate change impacted Odonata fauna by reducing the suitable habitats for the locally extinct specialist species (i.e. the decline of Mediterranean temporary ponds, Zacharias & Zamparas 2010). In Mauritania, we found a similar pattern of increasing average temperature in the last 50 years. Like for Tunisia, most of locally extinct Odonata were linked to temporary wetlands with abundant vegetation; given the semi-arid environment exposed to large rainfall variations and affected by a severe drought since the mid-1960s (Niang *et al.* 2008), it is likely that the habitat of these species may have suffered a strong reduction, thus resulting in local extinction or population decrease. For Sweden, Flenner and Sahlén (2008) recorded a clear turnover pattern from specialist to generalist species and concluded that climate change is the most probable cause. Our ecological analyses confirmed this pattern, with most of the extinct species being peri-arctic (Siberian-European distribution, *sensu* Vigna Taglianti *et al.* 1999), typically associated with moorlands, highland cool lentic waters with abundant vegetation, whereas new-

colonizers were generalist with extending range northward from their typical Central European distribution (Ott 2001, 2009).

In summary, a clear climate change (warming) occurred in our analysed regions during the last decades. We found three main patterns based on correlative evidence that may link some ecological traits of the selected species (i.e. those showing heterochronic distribution variation: contraction/extinction) to the positive variations in average temperatures:

i) Most generalist species have likely taken advantage of the warming processes that causes the loss of specific habitats (i.e. temporary wetlands, cool lentic waters) and the formation of new or altered habitats suitable for pioneer species (i.e. warm and intermittent pools; Zacharias & Zamparas 2010), whereas specialist species are more likely to go toward local extinctions;

ii) An overall pattern of northward expansion of the new colonizers from their southern distributions for two of the considered areas (Tunisia and Sweden);

iii) The Odonata communities inhabiting lentic waters are more prone to show species turnover than communities from lotic habitats. This is unexpected because the species inhabiting lentic waters are considered at lower extinction risk (Clausnitzer *et al.* 2009) partly due to their wider ecological preferences and higher dispersal capacity (Corbet 2004).

Although freshwater habitats, and the species therein included, are often considered as “particularly at risk” (Dudgeon *et al.* 2006, Dunn 2005, Naiman *et al.* 2006) due to anthropogenic pressures, relatively low level of threat to Odonata was found in the recent assessment by Clausnitzer *et al.* (2009). Unexpectedly, climate change was not considered among the main threats for Odonata (Clausnitzer *et al.* 2009) despite entire faunas are, or are expected to, disappearing jointly with their elective aquatic habitats (Erwin 2009, Strayer & Dudgeon 2010). In this scenario, our results provide new insights on the possible impact of climate change on Odonata fauna from large areas (i.e. countries).

### Potential shortcomings

We based our hypothesis of climate change-related species turnover on correlative evidence. However, a series of shortcomings possibly biasing our inferences may arise from (i) original faunistic data quality (especially for Mauritania dataset) and (ii) the attempt to categorize species into ecological groups. As for data quality, for Mauritania dataset, Ferreira et al. (2011) reported just 20 species in a country of over 1 million square km. Although Mauritania territory is desert for  $\frac{3}{4}$ , it is also characterized by areas of savannah, hills and is also adjacent to the Senegal River for hundreds of km. Moreover, Ferreira et al. (2011) stated: "the Mauritanian fauna still holds a good amount of potential surprise to stimulate the odonatologists"; hence there could be not enough elements to consider the absence of nine species as effectively caused by extinction. Regarding Tunisia, some of the species classified as extinct (Table 1) could be cases of past misidentification of the material of Gadeau de Kerville (1908), as pointed out by Jodicke et al. (2000), and the 10 species considered as new entries due to climate change could be also an effect of larger recent sampling activity. As for the ecological characterization of species, we based it mainly on literature data, and some biases could arise from species plasticity: i.e. one species could behave as lentic water dweller in some areas as well as lotic water dweller in others. In addition, we ignore the mechanistic processes occurring at the community level that could have contributed in shaping the observed pattern (i.e. species competitive interaction in a changing environment under climate change; see Cerini et al. 2019).

### Conclusion

Despite the issues listed above, our results on Odonata fauna turnover widely mirror those reviewed by Ott (2010) on the effect of climate change on range shifts and expansions. Overall, even if Odonata are currently the only insect group for which a global assessment of conservation status has been completed (Clausnitzer et al. 2009), insufficient information is available to robustly assess all the main threats affecting their status, and extensive new field surveys are required to test if major changes in fauna composition have occurred during the last decades (Flenner & Sahlén 2008). This holds especially true for high latitude and tropical areas where climate change has already produced a significant effect on species diversity and community composition (Walther et al. 2002, Moritz et al. 2008, Ott & Samways 2010, Freeman et al. 2018).

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